



Can intrafirm IT skills benefit interfirm integration and performance?



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ABSTRACT

This study develops a model to examine whether intrafirm, hard and soft IT skills have cross-boundary effects on interfirm collaboration and integration, thus leading to better supply-chain performance. A model with eight hypotheses was developed and tested using data collected from 250 Taiwanese manufacturing firms. Seven hypotheses find empirical support. The results show that intrafirm IT skills indeed can benefit interfirm collaboration and integration and thus, supply-chain performance. Our findings suggest that the value of skilled IT professionals for interfirm integration, supply-chain performance and collaborative relationships between firms is critical for realizing that value.

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1. Introduction

Over the last decade, firms have sought integration with their supply-chain partners to cope with uncertainty and to obtain better supply-chain performance [27,79,98]. Skilled business personnel certainly lie at the heart of interfirm collaboration for the integration. These business personnel from different departments, such as purchasing, play a boundary-spanner role by contributing their skills and knowledge across firms to help interfirm collaboration and integration [42]. Recently, advanced interorganizational systems (IOS) have profoundly changed the nature of interfirm operations in the supply chain [68,69,75]. Firms have recognized the importance of IOS and begun to use IOS to integrate their processes with those of their supply-chain partners to improve their supply-chain operations [68,69,75]. Prior studies also recognize that IOS integration can provide information visibility to mitigate the bullwhip effect [50], reduce the complexity of supply-chain activities [66], and promote flexibility to meet varying business demands [7]. However, it is not simple to integrate supply-chain partners using IOS. It requires the alignment of both processes and technology standards across firms [34] and demands that firms overcome the various difficulties caused by conflicting processes and standards [50].

To resolve these difficulties, skilled personnel from the cooperating firms must work together to address the conflicts.

In the past, business personnel from core departments, such as purchasing, manufacturing, and selling, typically served as the boundary-spanners, and their intense collaboration helped firms to achieve better interfirm integration [88], to improve goal alignment [66], to enable effective information flows, and to streamline logistics [66], thereby improving supply-chain performance. However, in the context of rapidly advancing IT, the role of IT personnel in enabling interfirm integration has become increasingly critical [68,69,75]. In interfirm operations, technology issues inevitably intertwine with business processes. Without skilled IT personnel and their involvement, firms have limited ability to tackle conflicting processes and technology standards [50], likely resulting in low levels of integration and mutual benefits. However, it remains unclear whether firms' internal IT skills can truly have a boundary-spanning effect when supply-chain partners attempt to integrate their information systems and processes. Consequently, whether and how intrafirm IT human capital, and IT skills in particular, can help interfirm collaboration and integration are issues of both academic and practical significance.

In supply chains, fulfilling collaborative objectives requires not only the participation of relevant personnel but also the application of their skills [73]. Although IT personnel tend to be more familiar with structured approaches and IT knowledge [72], recognized as hard skills, firms have increasingly search for business-literate IT personnel who can add value at a strategic level [44]. Although prior studies have recognized the importance

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